

LAB 3

1) Screenshot of passwd and shadow with description/notation of fields (10 points)

```
pulkit@desktop:~$ tail -n 10 /etc/passwd
speech-dispatcher:x:119:29:Speech Dispatcher,,,:/run/speech-dispatcher:/bin/false
nm-openvpn:x:120:126:NetworkManager OpenVPN,,,:/var/lib/openvpn/chroot:/usr/sbin/nologin
saned:x:121:128:/:/var/lib/saned:/usr/sbin/nologin
colord:x:122:129:colord colour management daemon,,,:/var/lib/colord:/usr/sbin/nologin
geoclue:x:123:130:/:/var/lib/geoclue:/usr/sbin/nologin
pulse:x:124:131:PulseAudio daemon,,,:/run/pulse:/usr/sbin/nologin
gnome-initial-setup:x:125:65534:/:/run/gnome-initial-setup:/bin/false
hplip:x:126:7:HPLIP system user,,,:/run/hplip:/bin/false
gdm:x:127:133:Gnome Display Manager:/var/lib/gdm3:/bin/false
pulkit:x:1000:1000:Pulkit Saxena,,,:/home/pulkit:/bin/bash
```

pulkit: It tells us the username

x: It is the placeholder for the password

1000: It tells us about the User ID(GID)

1000: It tells us about the Group ID(GID)

Pulkit Saxena: It tells us about the user's full name

/home/pulkit: It represents the home directory

/bin/bash: It represents the default shell

```
pulkit@desktop:~$ sudo tail -n 10 /etc/shadow
[sudo] password for pulkit:
speech-dispatcher:!:19101:0:99999:7:::
nm-openvpn:!:19101:0:99999:7:::
saned:!:19101:0:99999:7:::
colord:!:19101:0:99999:7:::
geoclue:!:19101:0:99999:7:::
pulse:!:19101:0:99999:7:::
gnome-initial-setup:!:19101:0:99999:7:::
hplip:!:19101:0:99999:7:::
gdm:!:19101:0:99999:7:::
pulkit:$y$j9T$kUqQR9ypldqEXr4i2Iy5F/$dDfsz.9X7L/JgbNTeIdiPNMV4ZXZPgPDiee2P2Epc.:20118:0:99999:7:::
```

pulkit: username

\$y\$j9T\$kUqQR9ypldqEXr4i2Iy5F/\$dDfsz.9X7L/JgbNTeIdiPNMV4ZXZPgPDiee2P2Epc.: Encrypted password

20118: The date on which the the password was changed recently

0: Minimum number of days left before password change

99999: Maximum number of days left before password change

7: The number of days before a user should start getting a warning message before password expiration.

2) Screenshot of passwd and shadow in middle of user account creation (10 points)

```
pulkit@desktop:~$ sudo tail -n 10 /etc/shadow
nm-openvpn:*:19101:0:99999:7:::
saned:*:19101:0:99999:7:::
colord:*:19101:0:99999:7:::
geoclue:*:19101:0:99999:7:::
pulse:*:19101:0:99999:7:::
gnome-initial-setup:*:19101:0:99999:7:::
hplip:*:19101:0:99999:7:::
gdm:*:19101:0:99999:7:::
pulkit:$y$j9T$KqQR9ypldqEXr4i2Iy5F/$dDfsz.9X7L/JgbNTEidiPNMV4ZXZPgPJDlee2P2Epc.:20118:0:99999:7:::
test_user:$y$j9T$Jg.3yrinkyfXXhVuhbQxE/$/LV7S0yWdooRVtQXZIZtxpexp.zZCFAZI76RjePjDf0:20132:0:99999:7:::
pulkit@desktop:~$ tail -n 10 /etc/passwd
nm-openvpn:x:120:126:NetworkManager OpenVPN,,,:/var/lib/openvpn/chroot:/usr/sbin/nologin
saned:x:121:128::/var/lib/saned:/usr/sbin/nologin
colord:x:122:129:colord colour management daemon,,,:/var/lib/colord:/usr/sbin/nologin
geoclue:x:123:130::/var/lib/geoclue:/usr/sbin/nologin
pulse:x:124:131:PulseAudio daemon,,,:/run/pulse:/usr/sbin/nologin
gnome-initial-setup:x:125:65534::/run/gnome-initial-setup:/bin/false
hplip:x:126:7:HPLIP system user,,,:/run/hplip:/bin/false
gdm:x:127:133:Gnome Display Manager:/var/lib/gdm3:/bin/false
pulkit:x:1000:1000:Pulkit Saxena,,,:/home/pulkit:/bin/bash
test_user:x:1001:1001::,/home/test_user:/bin/bash
pulkit@desktop:~$
```

3) Screenshot of new user's home directory contents (ls -la) and the output of the environment variables (echo \$USER \$SHELL) (5 points)

```
pulkit@desktop:~$ su - test_user
Password:
test_user@desktop:~$ ls -la
total 20
drwxr-x--- 2 test_user test_user 4096 Feb 13 00:10 .
drwxr-xr-x 4 root      root      4096 Feb 13 00:10 ..
-rw-r--r-- 1 test_user test_user  220 Feb 13 00:10 .bash_logout
-rw-r--r-- 1 test_user test_user 3771 Feb 13 00:10 .bashrc
-rw-r--r-- 1 test_user test_user  807 Feb 13 00:10 .profile
test_user@desktop:~$ echo $USER $SHELL
test_user /bin/bash
test_user@desktop:~$
```

4) Summarize the Key Differences between nologin and a locked password (5 points)

nologin	Locked password
This command the user to login	This command helps the user to change his password
It displays a message that an account is unavailable and then exits	It allows a normal user to his password for their account while the superuser has the authority to change password for any account. It can also change the validity period of this account.

5) Description of the effect of each of the (5) chmod commands

Ans)

Command	Effect
Chmod 777 filename	It grants the rights to read, write and execute to everyone(owner,group and others)
Chmod 700 filename	It grants the rights to read,write and execute only to the owner while the users and groups lose all their access.
Chmod u=rw filename	It grants the rights to read and write for the owner,while leaving the rights of the groups and users unchanged
Chmod go+x filename	It grants the permission to execute for groups and users while the rights of the owner remain unchanged.
Chmod a+w filename	It grants the permission to write for everyone (owner, groups and users)

6) Organized description of directory permissions and how affects contents (10 points)

Ans)

Read	Write	Execute
• It can be used to read the names of the files in a particular directory, however, it does not allow us to view its contents. • It can list the files by using the ls command but it cannot open them. • It can read the files if it has been granted with the permission "r".	• It allows us to create, delete or name files in a particular directory. • It can only modify the files if it has been granted with the permission "w". • It can be used to delete files even if it does not have the permission to do so.	• It allows us to access a directory and read the contents inside of it. • It can only allow us to enter the directory if it has been granted with the permission "x". • It allows us to access a file if we just know the filename just by having the permission as "--x".

7) Screenshot and description of /etc/shadow file - who can r/w/x and why it's set this way. (10 points)

```
pulkit@desktop:~$ mkdir my_test_dir
pulkit@desktop:~$ ls -ld my_test_dir
drwxrwxr-x 2 pulkit pulkit 4096 Feb 13 12:22 my_test_dir
pulkit@desktop:~$ touch my_test_dir/testfile.txt
pulkit@desktop:~$ ls -l /etc/shadow
-rw-r----- 1 root shadow 1460 Feb 13 00:53 /etc/shadow
pulkit@desktop:~$
```

Root(owner) has the right to read and write whereas shadow(group) can only read the file and users cannot do anything. The conditions have been set in such a manner as this file contains hash passwords and some other important data, so that's why only root(owner) has the right to modify the file in order to increase its security while shadow(group) can only read it and users cannot do anything.

8) Meanings of dig queries (10 points)

Query	What does the query result mean?
dig -t mx iastate.edu	It lists the mail servers handling email from iastate.edu, by showing the mail exchange (MX) records.
dig -t ns iastate.edu	It identifies authoritative DNS servers for iastate.edu, by displaying name server records.
dig -t soa iastate.edu	It shows the details about the primary DNS server by retrieving the Start of Authority Record (SOA)
dig -t aaaa iastate.edu	It returns the IPv6(AAAA) address, if it is available.
dig -t any iastate.edu	It returns any DNS records that are currently available.

**9) Screenshot of the (2) netstat outputs
(10 points)**

```
cpre230@ns1:~$ netstat -tl
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp      0      0 localhost:domain        0.0.0.0:*               LISTEN
tcp      0      0 0.0.0.0:ssh             0.0.0.0:*               LISTEN
tcp6     0      0 [::]:ssh                [::]:*                  LISTEN
cpre230@ns1:~$ netstat -tln
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp      0      0 127.0.0.53:53           0.0.0.0:*               LISTEN
tcp      0      0 0.0.0.0:22              0.0.0.0:*               LISTEN
tcp6     0      0 :::22                   :::*                     LISTEN
cpre230@ns1:~$ _
```

10) Complete the netstat flag table (10 points for all 3 correct meanings; 0 points for anything less)

Flag	Meaning
-t	It is used to display tcp connections
-l	It only shows the listening sockets
-m	It is used to display address in numeric format

11) Screenshot of the captured http traffic using Wireshark (5 points)

Wireshark interface showing captured HTTP traffic. The packet list displays several TLSv1.3 packets. The selected packet (No. 104773) is expanded, showing the Hypertext Transfer Protocol details. The packet bytes pane shows the raw data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length
1047...	70.565694478	42.49.30.201	199.100.16.100	TLSv1.3	460
1047...	70.565726533	42.49.30.201	199.100.16.100	TLSv1.3	121
1047...	70.577183497	199.100.16.100	42.49.30.201	TLSv1.3	680
1047...	70.577556787	199.100.16.100	42.49.30.201	TLSv1.3	97
1047...	70.577644803	42.49.30.201	199.100.16.100	TLSv1.3	97
1047...	70.601940106	199.100.16.100	42.49.30.201	TLSv1.3	417
1047...	70.601966135	199.100.16.100	42.49.30.201	TLSv1.3	1514
1047...	70.602010279	199.100.16.100	42.49.30.201	TLSv1.3	1514
1047...	70.602118932	199.100.16.100	42.49.30.201	TLSv1.3	1370
1047...	70.602121562	199.100.16.100	42.49.30.201	TLSv1.3	1514

Frame 104773: 97 bytes on wire (776 bits), 97 bytes captured (776 bits) on interface 0

Ethernet II, Src: VMware_86:81:87 (00:50:56:86:81:87), Dst: Cisco_31:01:b4 (08:00:0c:31:01:b4)

Internet Protocol Version 4, Src: 42.49.30.201, Dst: 199.100.16.100

Transmission Control Protocol, Src Port: 35954, Dst Port: 3128, Seq: 1561, Ack: 1561, Win: 0, Len: 0

Hypertext Transfer Protocol

0000 00 00 0c 31 01 b4 00 50 56 86 81 87 08 00 45 00 ...1...P V...E.

0010 00 53 61 4a 40 00 40 06 b8 98 2a 31 1e c9 c7 64 ...SaJ@.@...*1...d

0020 10 64 8c 72 0c 38 48 89 fc 83 04 fe 9c 7f 80 18 ...d.r.8H.....

0030 01 f5 21 08 00 00 01 01 08 0a 73 07 55 cb 71 53 ...!.....s.U.qS

Hypertext Transfer Protocol: Protocol Packets: 105173 · Displayed: 1111 (1.1%) Profile: Default

12) Screenshot of the captured icmp traffic using tcpdump (5 points)

```
eq 25223, length 64
13:19:40.560934 IP 72.208.195.200 > 72.208.195.254: ICMP echo request, id 9, seq 6566, length 64
13:19:40.561075 IP 72.208.195.254 > 72.208.195.200: ICMP echo reply, id 9, seq 6566, length 64
13:19:41.423821 IP 84.56.33.200 > 199.100.16.100: ICMP 84.56.33.200 udp port domain unreachable, length 102
13:19:41.475268 IP 72.208.195.201 > 72.208.195.200: ICMP echo request, id 2, seq 25224, length 64
13:19:41.475346 IP 72.208.195.200 > 72.208.195.201: ICMP echo reply, id 2, seq 25224, length 64
13:19:41.584856 IP 72.208.195.200 > 72.208.195.254: ICMP echo request, id 9, seq 6567, length 64
13:19:41.585067 IP 72.208.195.254 > 72.208.195.200: ICMP echo reply, id 9, seq 6567, length 64
13:19:42.499271 IP 72.208.195.201 > 72.208.195.200: ICMP echo request, id 2, seq 25225, length 64
13:19:42.499345 IP 72.208.195.200 > 72.208.195.201: ICMP echo reply, id 2, seq 25225, length 64

pulkit@desktop:~/Desktop$ ping -c 5 42.49.30.200
PING 42.49.30.200 (42.49.30.200) 56(84) bytes of data.
64 bytes from 42.49.30.200: icmp_seq=1 ttl=64 time=0.441 ms
64 bytes from 42.49.30.200: icmp_seq=2 ttl=64 time=0.169 ms
64 bytes from 42.49.30.200: icmp_seq=3 ttl=64 time=0.180 ms
64 bytes from 42.49.30.200: icmp_seq=4 ttl=64 time=0.196 ms
64 bytes from 42.49.30.200: icmp_seq=5 ttl=64 time=0.184 ms

--- 42.49.30.200 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4091ms
rtt min/avg/max/mdev = 0.169/0.234/0.441/0.103 ms
pulkit@desktop:~/Desktop$
```